

■ ECOSYNAPSE™ MASTER PLATFORM MANUAL

Complete Technical & Non-Technical Guide: Architecture, Workflows, Algorithms, Hardware & User Operations

PART 1: NON-TECHNICAL MANUAL & USER OPERATIONAL GUIDE

1.1 System Vision & Value Proposition

EcoSynapse is an intelligent community waste management platform that transforms traditional waste disposal into a smart, data-driven circular economy. By combining touchless smart bin hardware, AI computer-vision segregation verification, and algorithmic rewards, EcoSynapse motivates residents to segregate waste correctly at source while helping facility managers and municipality collectors optimize collection routes.

1.2 Resident User Guide (How to Earn & Redeem EcoPoints)

Step 1: Logging in to the User Portal

Residents log in using their apartment credentials (e.g. sriram@ecosynapse.com / EcoSynapse2026!) or click **Continue as Guest**. Upon logging in, residents land on their personal **User EcoPoints** dashboard.

Step 2: Approaching & Disposing Waste at the Smart Bin

- **Touchless Lid Opening:** Stand in front of your assigned smart bin (e.g. Greenwood Block B Lobby - BIN-102). The Time-of-Flight (ToF) proximity sensor automatically lifts the lid.
- **Segregation:** Deposit dry recyclables (paper, plastic, glass, cans) into the **Dry Compartment** and food scraps into the **Wet Compartment**.
- **Instant Verification:** The camera and sensors inspect the deposit in real time. If verified dry waste is detected, your account is credited with **+10 EcoPoints**.

Step 3: Understanding Your EcoScore & User Tier

Your monthly **EcoScore** (0 to 100) determines your **User Tier** and point multiplier:

- ■ **Green Tier (< 50 Score):** 1.0x Multiplier (Standard point earning).
- ■ **Silver Tier (50 – 69 Score):** 1.1x Multiplier (+10% extra bonus points).
- ■ **Gold Tier (70 – 89 Score):** 1.25x Multiplier (+25% extra bonus points).
- ■ **Platinum Tier (≥ 90 Score):** 1.5x Multiplier (+50% extra bonus points & AGM certificate recognition).

Step 4: Redeeming Vouchers & Perks

Residents can exchange accumulated EcoPoints in the **Redeem Marketplace** for real rewards:

- ■ **100 Instant UPI Cash Voucher** (300 Pts)
- ■ **Plant a Sapling in Society Park** (100 Pts)
- ■ **Public Transport Metro / Bus Pass** (200 Pts)
- ■ **Electricity & Utility Bill Credit** (350 Pts)
- ■ **200 Apartment Maintenance Fee Voucher** (400 Pts)

1.3 Sanitation Collector Operations Manual

Role Access: Sanitation drivers log in under the **Collector Portal** role.

- **■ Live Priority Dispatch:** When a smart bin reaches 80% capacity or flags a liquid leak/contamination, it automatically appears at the top of the Collector App with a red pulsing badge.
- **■ Voice Alert Notifications:** Collectors tap **Play Voice Alert** to hear automated speech synthesis announcing exact bin locations (*Attention Driver: Greenwood Block B is 93% full*).
- **■ Marking Bins Emptied:** Upon emptying a smart bin node, the driver taps **Mark Emptied**. This resets the fill level to 0%, clears fault alerts, and logs the route completion timestamp.

PART 2: TECHNICAL MANUAL, ARCHITECTURE & HARDWARE

2.1 Full Technology Stack Specification

Component Layer	Technology / Framework	Implementation Detail
Frontend Framework	Next.js 14 (App Router) + React 18	Server and Client Component rendering, responsive design, fast page loads.
Language & Styling	TypeScript + Tailwind CSS	Strict typing, zero runtime CSS overhead, custom theme dark/light modes.
Iconography & UI	Lucide React Icons	High-performance SVG icon components.
Database Storage	Supabase PostgreSQL Database	Relational tables (users, bins, system_events, rewards) with SQL RLS.
Authentication	Supabase Auth + Local Fallback	Hybrid authentication supporting JWT sessions and instant offline demo fallback.
AI Assistant LLM	Google Gemini API (@google/genai)	EcoBot AI assistant with natural language understanding of waste telemetry.
Audio Speech System	Web Speech Synthesis API	Browser-native voice prompt announcements (window.speechSynthesis).
Edge Hardware	ESP32 + ESP32-CAM Microcontroller	Wi-Fi telemetry, image capture, weight load cells, moisture sensor grid.

2.2 Edge Layer Hardware Telemetry Specs

- **ESP32 Microcontroller:** Dual-core LX6, 2.4 GHz Wi-Fi + Bluetooth LE, low power deep-sleep state.
- **ESP32-CAM Module:** OV2640 camera sensor capturing 1600x1200 resolution snapshots upon waste disposal.
- **Dual Ultrasonic Sensors (HC-SR04):** Mounted on lid ceiling to continuously measure dry and wet waste levels.
- **Load Cell & HX711 Amplifier:** Measures weight increment (0.01 kg resolution) added per deposit.
- **Capacitive Moisture Sensor Grid:** Detects wet organic contamination placed inside dry compartment (>65% threshold).
- **Bottom Sump Sensor:** Detects liquid leachate pooling at bin base.
- **ToF Distance Sensor (VL53L0X):** Infrared laser proximity sensor detecting hand gesture for touchless lid opening.

PART 3: WORKFLOW FLOWCHARTS & MATHEMATICAL ALGORITHMS

3.1 Master System Data Flowchart (Visual Grid Diagram)

Step / Node	Action / Component	System Logic & Decision Branch
1. Waste Deposit	Resident approaches Smart Bin	ToF proximity sensor detects hand gesture → ESP32 opens lid automatically.
2. Intake Sensing	ESP32-CAM + Moisture + Weight	Captures image, measures added weight (kg), and reads moisture grid wetness %.
3. AI Verification	Segregation Score Engine	If Moisture < 65% and Vision = Dry → Verified Disposal (+10 EcoPoints). If Moisture > 65% in Dry Bin → Flag CONTAMINATION_ALERT.
4. Telemetry Sync	Cloud API Gateway (/api/bins)	WebSockets syncs updated bin capacity % and logs system audit event.
5. Route Dispatch	Fill Level Evaluation (≥ 80%)	If overallFill ≥ 80% → Escalate to Collector Priority Dispatch + Voice Alert. Collector clicks 'Mark Emptied' → Reset capacity to 0%.

3.2 Mathematical Formulas & Algorithmic Engines

1. EcoPoints Transaction Formula (EP)

EP = (SA × 40%) + (RP × 25%) + (WR × 20%) + (CS × 10%) + (CB × 5%)

- **SA (Segregation Accuracy Score, 0-100):** AI vision confidence + moisture verification score.
- **RP (Recycling Participation Score, 0-100):** Weight & frequency score for dry recyclable deposits.
- **WR (Waste Reduction Score, 0-100):** Historical reduction trend in overall waste generation.
- **CS (Consistency Streak Score, 0-100):** Daily consecutive segregation streak (+2 to +10 pts).
- **CB (Correct Disposal Bonus, 0-100):** Bonus for hazardous/e-waste segregation.

2. EcoScore User Tier Multiplier Engine

Final EcoPoints = EP × Tier Multiplier

3. Predictive Bin Overfill Velocity Formula

$$\Delta \text{ FillPerHr} = (\text{OverallFill_t2} - \text{OverallFill_t1}) / (\text{t2} - \text{t1})$$

$$\text{Predicted Hours} = \text{Max}(0.1, (100 - \text{CurrentOverallFill}) / \Delta \text{ FillPerHr})$$

When OverallFill ≥ 90%, predicted hours drops below 1.0 hour, triggering automatic route dispatch escalation.

PART 4: SECURITY, RBAC & TROUBLESHOOTING MANUAL

4.1 Security & Access Control Enforcement

To prevent unauthorized access, EcoSynapse enforces strict role-based route guards in Next.js:

- **Resident Protection:** Resident accounts (activeRole === 'user') are strictly restricted to the User EcoPoints dashboard. The Collector App tab, Admin Fleet tab, and **Mark Emptied** action buttons are hidden and inaccessible.
- **Collector Protection:** Only verified Sanitation Driver accounts (activeRole === 'collector') or Administrators can execute bin emptying commands.
- **Hybrid Auth Resiliency:** If cloud Supabase authentication is offline, the system seamlessly uses secure local token validation so community services never experience downtime.

4.2 System Maintenance & Troubleshooting Guide

Symptom / Issue	Probable Cause	Resolution / Action Step
Login error on demo email	Supabase Auth email unconfirmed	Fixed in v2.0. Hybrid auth fallback automatically verifies demo accounts (sriram@ecosynapse.com).
Collector tab missing in header	Logged in as Resident role	Log out and sign in under Collector Portal role or Admin Manager role.
Voice alerts silent	Browser audio muted or unsupported	Click 'Play Voice Alert' button to enable SpeechSynthesis browser permissions.
Contamination warning on Dry Bin	Moisture grid level > 65%	Inspect dry compartment; remove wet organic items and wipe capacitive sensors.

Conclusion & Support: EcoSynapse v2.0.0 is fully built, compiled, and production ready. For technical inquiries, contact support@ecosynapse.com.